

Simulate SMARTPHONE SCREEN LOCK/UNLOCK APPLICATION

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An Android phone allows itself to be locked/unlocked using a password, pattern on a grid of nine dots, fingerprint or facial recognition. An Android phone using a password needs a sequence of four digits or numbers to unlock the screen. This article describes the design, testing and working of the smartphone screen lock/unlock application using Simulink and Stateflow toolboxes of MATLAB R2014a software.

Working

The design of the phone screen lock/unlock application, as given in Fig. 3, contains paths S and E for the password (1 3 2 5). Path S is $S_0 \rightarrow S_1 \rightarrow S_2 \rightarrow S_3 \rightarrow S_4$, whereas path E is $E_1 \rightarrow E_2 \rightarrow E_3 \rightarrow E_4$. All Android phones can be unlocked by entering the four magic digits set as password by users.

The screen lock/unlock application is designed in such a way that a correct four-digit password entered by the user chooses path S ($S_0 \rightarrow S_1 \rightarrow S_2 \rightarrow S_3 \rightarrow S_4$) to unlock the phone. However, entering a wrong password causes the transition jump from path S to path E, leading to the failure of the password and the screen to remain locked.

When a wrong password is entered, the phone remains locked. Some examples are given below:

1. For password 9 3 2 5, the phone takes path $S_0 \rightarrow E_1 \rightarrow E_2 \rightarrow E_3 \rightarrow E_4$ and remains locked.
2. For password 1 4 2 5, the phone takes path $S_0 \rightarrow S_1 \rightarrow E_2 \rightarrow E_3 \rightarrow E_4$ and remains locked.
3. For password 1 3 5 5, the phone takes path $S_0 \rightarrow S_1 \rightarrow S_2 \rightarrow E_3 \rightarrow E_4$ and remains locked.
4. For password 1 3 2 1, the phone takes path $S_0 \rightarrow S_1 \rightarrow S_2 \rightarrow S_3 \rightarrow E_4$ and remains locked.

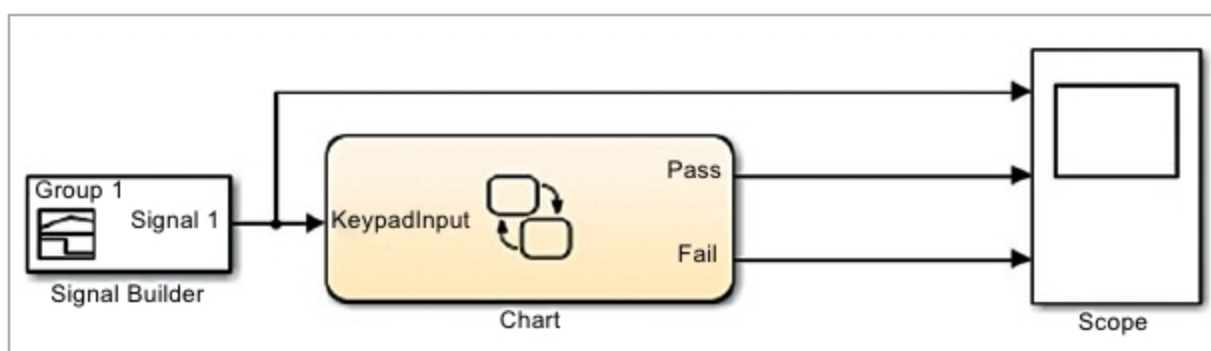


Fig. 1: The Stateflow model of the application

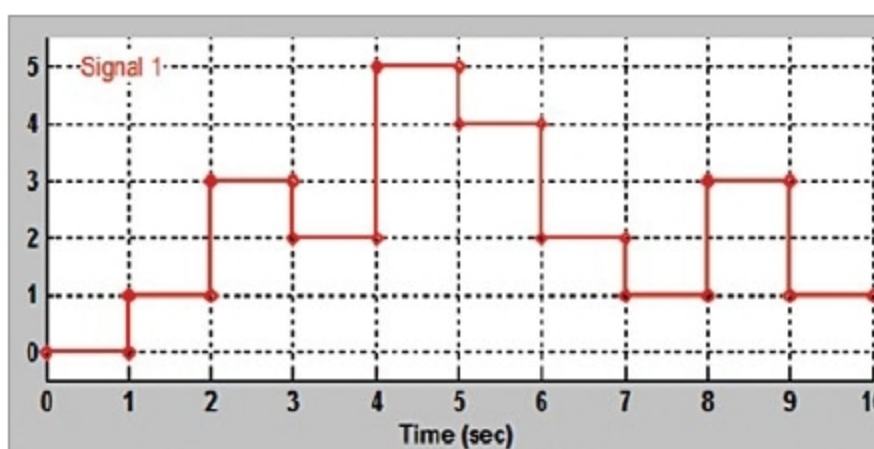


Fig. 2: Input waveform for the stored password

The phone screen lock application presented here can also be used in other places where four-digit authentication is required, for, say, unlocking a digital door lock, drawing cash from an ATM machine and so on.

Modelling simulation

Steps for creating the phone screen lock application using MATLAB software are given below.

1. Open MATLAB software and enter simulink command to open Simulink Library Browser.
2. Open a new blank model for creating the screen lock application by selecting File→New→Model from Simulink Library Browser.
3. Take three blocks—Signal Builder, Chart and Scope—as shown in Fig. 1 from Simulink Library Browser to build the model. First, click on Simulink toolbox in Simulink Library Browser and then

Sources library to select Signal Builder block. Then, take Scope block from Sinks library in Simulink toolbox. Last, take Chart block from Stateflow toolbox.

4. Pick, drag and place the three blocks—Signal Builder, Chart and

Scope—from Simulink Library Browser into Model.

5. Double-click on the chart to open the internal structure or state machine of the simulation. A new window will open, as shown in Fig. 3. States and transitions are the key graphical elements used for building the logic of the screen lock application inside the chart.

6. Graphical elements (states, transitions, etc) needed to create the logics are available on Object Palette inside the chart.

7. Create the logic inside the chart using the states and drawing transitions, as given in Fig. 3. Right-click anywhere inside the chart, select Properties option. Next, choose Update Method as Discrete, and enter Sample Time as 1.

8. After creating the logic, select View→Model Explorer from the menu to add input and output to the chart.